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FIRST SEMESTER B.C.A. (REVISED NEP) DEGREE
EXAMINATION, JANUARY/FEBRUARY 2025
DISCRETE MATHEMATICAL STRUCTURES (DSC – 5)

Time : 3 Hours] [Max. Marks : 80

***Instruction** : Answer **all** the Sections based on **internal** choices.*

SECTION – A

I. Answer **all** of the following questions. **Each** carries **2** marks.

1) Explain any two logical connectives with truth table.
2) Define power set with an example.
3) Find the number of permutation of word "BANANA".
4) Define quantifiers. Mention its types.
5) Find the number of ways in which 7 books can be selected out of 10 books. 120.
6) Define recursive algorithm.
7) Define function. Mention any two functions.
8) What is equivalence relation ?
9) Define graph. Mention graph terminologies.
10) What is minimum spanning tree ? **(10×2=20)**

SECTION – B

II. Answer **any six** questions. **Each** carries **5** marks.

11) Prove that $[[(p\rightarrow q) \wedge (p\rightarrow r)]\rightarrow(p\rightarrow r)]$ is tautology.
12) Define set. Explain all the set operation with Venn diagram.
13) Out of 6 boys and 4 girls a committee of 6 to be form. In how many ways can this done if committee contains
   i) 2 girls
   ii) Atleast 2 girls ?
14) Obtain the Co-efficient of x^9y^3 in the expansion $(2x – 3y)^{12}$. 220

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15) The Fibonacci number are defined recursively by $F_0 = 0$, $F_1 = 1$ and $F_n = F_{n-1} + F_{n-2}$ $n \geq 2$. Find the F6.

16) Prove that by Mathematical induction $1^2 + 2^2 + 3^2 + \ldots\ldots + n^2 = \frac{n(n+1)(2n+1)}{6}$.

17) Explain Dijkstra's algorithm.

18) Explain Graph coloring with example.

(6×5=30)

SECTION – C

III. Answer **any three** questions. **Each** carries **10** marks.

19) Explain the rules of inference.

20) a) State and prove generalised pigeonhole principle with example.

b) Certain question paper contains two parts A and B each containing 4 questions. How many different ways a student can answer 5 questions by selecting atleast 2 questions from each part ? 10

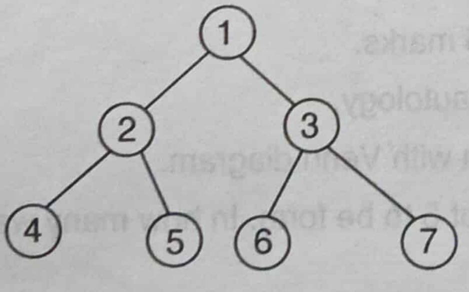
21) a) Find the zero-one matrix of the transitive closure of the relation R where :

$$M_R = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 1 & 0 \end{bmatrix}$$

b) Define relation. Explain the properties of relation.

22) a) Explain the different types of graph.

b) Define tree traversal. Traverse the following tree with inorder, preorder and postorder traversal.



(3×10=30)